

# Rows Columns and Data Types

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Classify records for Rows columns and schemas. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Rows columns and schemas. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Design a schema independently

For a fictional shelf audit, define label and book\_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-input case.

## 6. Solve the given case

Fields are Title, Pages and Borrowed; a row contains River, ninety and yes.

Assign intended types and a repair.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A table row is book\_id B7, pages 84, available true. Name the type of each field and explain why pages stored as eighty-four complicates calculation.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 7 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Store one book per row and keep reader preferences in a separate record.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

A row contains title='Sky', pages='120', available='false'. Explain why treating nonempty text as true can misread availability. Propose explicit conversions and a rejected-input case.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 State a schema

List the required fields, types and duration boundaries for the worked table.

**My reasoning and evidence**

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**13. W2 Classify records**

Classify minutes values 0, 60, -1 and 61 under this schema.

**My reasoning and evidence**

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**14. W3 Repair a unit problem**

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

**My reasoning and evidence**

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## 15. W4 Design a schema independently

For a fictional shelf audit, define label and book\_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

### My reasoning and evidence

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## 16. W5 Separate validity and truth

A row says Reading, 12 and passes. Does that establish the session really lasted 12 minutes?

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

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**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Rows Columns and Data Types

A fictional approved book catalogue has three fields per row.

Schema: id is nonempty text; pages is an integer 1–300; illustrated is Boolean True or False.

Rows A={id:"B7",pages:90,illustrated:True}; B={id:"B8",pages:"90",illustrated:False};

C={id:"",pages:80,illustrated:True}.

## Rows Columns and Data Types

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Column contract

Field names identify values; order is not the only guide.



#### 2. Validate A

B7 is text, 90 is allowed integer, True is Boolean: valid.



#### 3. Validate B

Quoted "90" is text, so pages fails the integer contract.



#### 4. Validate C

Empty id fails even though its other two values have allowed types.

### CHANGE ONE THING

Change B's pages from "90" to integer 90 using an approved repair.

Explain the changed result using the matching lesson.

Which two rows originally fail, and do they fail for the same reason?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Publish the recommender's row schema and validation results before ranking books, preserving repair provenance.